

Addressing Impostor Phenomenon in Collaboration Between User Experience Designers and Software Developers

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Abstract

This study examines the impact of Impostor Phenomenon (IP) on collaboration between UX designers and developers, an area underexplored in human-computer interaction and software development. Qualitative interviews reveal that IP disrupts collaboration and productivity by fostering procrastination, overwork, anxiety, and self-doubt. It also exacerbates communication barriers and creates a reluctance to share ideas, further hindering effective teamwork. Grounded in five key themes, this research identifies critical challenges posed by IP and lays the foundation for the development of design-driven interventions. Future work will focus on participatory workshops to co-create solutions aimed at mitigating IP's effects and fostering more effective interdisciplinary collaboration.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Software and its engineering**; • **Software creation and management**; • **Collaboration in software development**;

Keywords

Design Thinking, Software Development, Collaboration

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1 Introduction

This research explores the impact of Impostor Phenomenon (IP) on the collaboration between user experience (UX) designers and developers, as effective team dynamics are essential for successful software development and user experience. UX designers focus on creating interfaces and user experiences, while developers implement these designs into functional software systems or websites [32, 43]. Collaboration between these roles, however, often faces challenges such as differences in terminology, workflows, and team

structures, which can hinder communication and alignment in project goals.

IP, a psychological phenomenon, involves individuals doubting their achievements and fearing exposure as frauds, even when successful [12]. This state often leads to fear of feedback, reluctance to share ideas, overwork, and procrastination, which disrupt communication and decision-making processes [13, 46, 47, 50, 51]. In collaborative environments, these tendencies can exacerbate existing challenges, causing UX designers and developers to hesitate in sharing ideas or providing feedback, ultimately hindering teamwork and project outcomes. This research hypothesizes that IP not only affects individuals but also amplifies collaboration barriers in interdisciplinary teams.

While existing research has extensively studied the psychological impacts of IP, limited attention has been given to its role in interdisciplinary collaboration, particularly in the context of UX designers and developers. Furthermore, current intervention strategies [44, 48] primarily focus on psychological approaches such as counseling or reflective journaling, with little emphasis on design-driven methods. Communication challenges are also a critical factor in collaboration. Leiva et al. [35] highlight issues such as "information loss" and "technical constraints," which emphasize the importance of effective communication in interdisciplinary teams. These challenges intersect with the effects of IP, as individuals experiencing self-doubt may struggle to articulate ideas or negotiate solutions effectively. This intersection underscores the need for interventions that address both psychological and collaborative barriers.

Initial thematic analysis from semi-structured interviews has identified key challenges, including communication barriers, emotional stress, and productivity impacts. These findings highlight specific opportunities where design interventions could mitigate the negative effects of IP, such as tools to facilitate open communication and reduce over-preparation. Building on these findings, participatory design workshops have been conducted to collaboratively explore actionable solutions. The ongoing analysis of workshop data aims to refine these interventions and ensure their relevance to real-world practice.

To address these challenges, this study is guided by the following research questions:

RQ1: How do experiences of Impostor Phenomenon (IP) between UX designers and developers affect their collaboration, and what needs arise from these experiences?

RQ2: How might the impact of Impostor Phenomenon (IP) on UX designers and developers working together be mitigated through design to improve collaboration?

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This study contributes to the understanding of how Impostor Phenomenon (IP) affects interdisciplinary collaboration by identifying its specific effects on communication, emotional well-being, and productivity in UX design and development teams. It introduces a novel approach by integrating design thinking with participatory design to co-create interventions that foster open communication, reduce collaboration barriers, and enhance team dynamics.

By addressing the psychological and collaborative challenges faced by UX designers and developers, this research aims to provide innovative strategies for improving team dynamics and productivity. Future work will involve designing and testing a desktop software prototype as an intervention to mitigate the effects of Impostor Phenomenon (IP). This prototype will be evaluated in real-world team settings to explore its effectiveness and scalability across diverse organizational contexts. This approach seeks to offer practical solutions that enhance both collaboration and user experience outcomes.

2 Related work

2.1 Impostor Phenomenon (IP) and Its Workplace Impacts

Impostor Phenomenon (IP) is a psychological pattern where individuals doubt their accomplishments and fear being exposed as frauds, despite evident successes [12]. Originally identified among high-achieving women, IP is now recognized as affecting individuals across genders and demographics [10, 18, 46, 49]. Research links IP to reduced job satisfaction, emotional exhaustion, and increased burnout risk, as well as compromised creativity and organizational commitment [22, 25, 27, 28, 53].

Professionally, IP often leads individuals to over-focus on immediate tasks due to fear of exposure, neglecting broader responsibilities and career planning [53]. Emotional impacts such as shame and anxiety further impair job performance, teamwork, and communication [22, 25]. This is particularly evident during career transitions or in competitive environments, where social comparisons and external pressures amplify feelings of inadequacy [5, 14, 16, 21, 26, 31, 42]. The Impostor Cycle describes a recurring pattern of overexertion, temporary relief, and renewed self-doubt, perpetuating these challenges [13].

Interventions for IP include both individual and group-based approaches. Group interventions, such as workshops and group psychotherapy, foster shared understanding and reduce feelings of isolation [12, 23, 40]. Individual approaches, such as counselling and reflective journaling, help address personal challenges and develop coping strategies [1, 15, 55]. Despite these advances, there is limited exploration of design-focused strategies that address IP's impact on professional collaboration.

2.2 Collaboration Between UX Designers and Developers

Collaboration between UX designers and developers is essential in the software development lifecycle. Designers focus on creating user interfaces and experiences, while developers implement these designs using front-end technologies [43]. Successful collaboration requires leveraging the expertise of both roles to improve software

quality and user satisfaction [6, 19, 35]. However, significant challenges often arise due to differences in workflows, terminology, and goals [4, 11, 29, 35].

Agile Software Development prioritizes adaptability and rapid delivery, while UX design emphasizes minimizing changes to meet user needs, leading to potential conflicts [4]. Studies highlight issues such as "information loss" during design handoffs, overlooked "edge cases," and "technical constraints" that necessitate redesigns [35, 36]. Poor communication, including vocabulary mismatches and a lack of mutual understanding, exacerbates these challenges [2, 30, 32, 39, 54].

Empathy also plays a crucial role in collaboration. Designers often report feeling marginalized when developers perceive design as secondary to technical work, resulting in resistance to implementing designs and diminished team cohesion [39]. Addressing these challenges requires fostering mutual awareness, inclusive participation, and stronger communication strategies [2, 17, 30].

2.3 Addressing the Gap: IP in UX-Developer Collaboration

While IP is widely studied in individual and organizational contexts, its impact on collaboration between UX designers and developers remains underexplored. Fear of evaluation, a key manifestation of IP, can discourage individuals from sharing ideas and feedback, stifling team creativity and progress [5, 12, 47]. Additionally, IP-related perfectionism and procrastination can hinder timely decision-making and project execution [33, 46, 52].

Despite the growing recognition of these challenges, existing interventions largely focus on psychological frameworks, with limited emphasis on addressing IP through design-driven strategies. Recent studies suggest design methods could play a vital role in mitigating IP's negative effects. For example, UX prototypes have been used to reduce IP during onboarding, enhancing confidence and cohesion in remote teams [34, 37, 38]. However, the application of these strategies to UX-developer collaboration is still in its infancy.

This research seeks to bridge this gap by investigating how IP influences collaboration dynamics and by developing design-based interventions that foster communication, reduce collaboration barriers, and enhance team productivity. By addressing the interplay between psychological and collaborative challenges, this study contributes to a deeper understanding of IP in interdisciplinary teams and offers actionable solutions to improve professional practice.

3 Method

This research employs the design thinking framework, which is particularly suited for addressing complex collaboration challenges between UX designers and developers. Based on the Hasso Plattner Institute of Design's model, the process consists of five key phases: empathize, define, ideate, prototype, and test. These phases inform one another, creating a systematic and iterative approach to solving Impostor Phenomenon (IP)-related issues in team collaboration.

3.1 Empathize: Semi-Structured Interviews

The empathize phase investigates how IP affects collaboration between UX designers and developers. Semi-structured interviews

Table 1: Demographic characteristics of participants

Participants	Gender	Job Role	Experiences	Educational Background
UXD1	Female	UX Researcher	5-10 years	PhD in Design
UXD2	Male	UX Designer	2-5 years	Diploma in UX Design
UXD3	Female	UX Designer	2-5 years	Master's in Industrial Design
UXD4	Male	UX Lead	10+ years	Master's in Human-Computer Interaction
UXD5	Female	UX Designer	2-5 years	Bachelor's in Information Systems and Marketing
UXD6	Male	Service Designer	2-5 years	Master's in User Experience Design
UXD7	Male	UX Designer	2-5 years	Master's in User Experience Design
DEV1	Female	Web Developer	2-5 years	Bootcamp Training in Software Development
DEV2	Male	Software Developer	2-5 years	Master's in Software Development
DEV3	Male	Front-end Developer	10+ years	Training Course in Computer Programming
DEV4	Male	Web Developer	10+ years	Diploma in Web Development
DEV5	Male	Front-end Developer	2-5 years	Bachelor's in Engineering (Honors)
DEV6	Female	Front-end Developer	2-5 years	Bachelor's in 3D Animation
DEV7	Female	Software Developer	10+ years	Graduate Diploma in Information Technology

^a UXD refers to professionals working in User Experience-related roles. DEV represents individuals engaged in development roles. Educational background is based on the highest degree most relevant to the participant's current role.

were conducted to explore participants' experiences and perceptions of IP within their professional collaborations [45]. This method provides rich, qualitative insights into how IP manifests in professional settings and impacts team dynamics.

3.1.1 Participant Recruitment and Demographics. Participants were recruited through purposive sampling using online communities, LinkedIn, and professional networks. Additionally, snowball sampling was employed, where existing participants referred colleagues who met the eligibility criteria. Eligible participants included UX designers and developers with at least one year of collaborative experience. A total of 14 participants were involved, comprising 7 UX designers and 7 developers from diverse professional backgrounds and geographic locations.

All participants self-reported having experienced Impostor Phenomenon (IP) in their work, based on Clance and Imes' [12] definition of IP as persistent self-doubt and fear of being exposed as a fraud despite evident success. This criterion was used to ensure that the study captured perspectives directly relevant to the research objectives. Interviews were conducted virtually or face-to-face, depending on participant preference, and were audio-recorded and transcribed using Otter AI with informed consent to ensure data integrity and confidentiality [24, 41].

3.1.2 Interview Procedure. The study has received prior ethical approval. Each interview lasted approximately 30-60 minutes and followed a semi-structured format. Key discussion points included:

- Participants' understanding and experiences of Impostor Phenomenon.
- Challenges faced during collaboration with counterparts.
- Perceptions of how collaboration dynamics could be improved.

This approach enabled participants to share specific examples of IP affecting their teamwork, providing detailed and actionable insights.

3.2 Data

We utilized Otter AI (otter.ai) to transcribe all interview recordings, ensuring accurate representation of participant responses. The transcribed data was then imported into NVivo, a qualitative data analysis software, to facilitate systematic thematic analysis. Following Braun & Clarke's framework [7–9], the analysis process was iterative and reflexive, focusing on identifying meaningful patterns related to Impostor Phenomenon (IP) and its impact on collaboration between UX designers and developers.

The analysis involved reviewing transcripts to ensure familiarity with the data and generating initial codes within NVivo 15 (A qualitative analysis software). These codes were iteratively grouped into potential themes, which were further refined through collaborative discussions among researchers. NVivo's features, such as query tools and visualizations, were used to organize and explore the data systematically, ensuring that the themes aligned with the research objectives. Differences between participant roles were also examined to capture both shared and role-specific insights.

This analysis provides critical insights into how IP affects collaboration, forming the basis for defining challenges and designing targeted interventions in subsequent research phases.

4 Findings

4.1 Theme 1: Bridging Understanding and Cross-Disciplinary Collaboration to Mitigate Impostor Phenomenon

Misalignments in understanding, priorities, and roles between UX designers and developers exacerbate Impostor Phenomenon (IP) and hinder collaboration. Participants emphasized fostering cross-disciplinary understanding as a key strategy to alleviate these challenges.

Differences in team priorities often created tension. For example, a designer highlighted how unrealistic expectations from developers amplified feelings of inadequacy: "They just wanted this instant;

[Name] will know it's in her head. Design is not just in your head, doesn't it?" (UXD1)

Participants also noted that stereotypes and technical jargon strained team dynamics, further fueling self-doubt and misunderstandings: "Designers saying that, developers like geeks or they don't really enter a vibe to it." (DEV2)

"Because they tend to use technical jargon, assuming everyone understands it." (UXD7)

To address these issues, participants suggested regular meetings and collaborative discussions to bridge gaps and improve mutual understanding: "Maybe have a meeting once in a while and then maybe the designer can explain the reasoning or why they make the application look like this... Then the OS developer will understand more and maybe try to understand each other better." (DEV6)

4.2 Theme 2: Communication and Interaction as a Foundation for Mitigating Impostor Phenomenon

Effective communication is essential for mitigating the challenges posed by Impostor Phenomenon (IP) in collaborative environments. Participants described how IP exacerbates communication barriers, fosters self-doubt, and undermines team dynamics.

Fear of feedback was a prominent issue, with participants often misinterpreting constructive criticism as personal failure. A developer shared: "But it's how I take that information or that feedback, and move forward with it. Do I take it and go down the route of Oh god, I'm no good at this?" (DEV4)

Similarly, a designer expressed uncertainty when receiving critiques: "When they critique the visuals, I often feel torn, questioning which aspect isn't satisfying and how I could improve it." (UXD3)

Participants emphasized the importance of transparency and direct feedback mechanisms. A developer suggested openly discussing challenges to foster mutual support: "I think the biggest thing is to raise your hand and say look, I'm struggling here. And I think it's because of these reasons." (DEV3)

Designers highlighted involving developers earlier in the design process to reduce misunderstandings and improve collaboration: "I would like developers to be more involved in this process, even if it might mean more feedback from them." (UXD7)

These findings underscore the need for open communication, shared understanding, and feedback practices to alleviate IP's impact and enhance team collaboration.

4.3 Theme 3: Emotional and Behavioral Impact of Impostor Phenomenon on Collaboration

The Impostor Phenomenon (IP) disrupts collaboration by fostering self-doubt, hesitation, and avoidance. These behaviors weaken team cohesion and limit individual contributions.

A developer described suppressing their ideas due to self-doubt: "So sometimes I can kind of push my opinions down or not bring them up, just because of not feeling like I have the same qualification as the designers themselves." (DEV7)

Similarly, a designer highlighted how fear of rejection hindered their confidence in sharing ideas: "It's hard to tell what's better right away. And when they suggest something, it's difficult to refuse. You start wondering, 'Should I go with their idea or mine?'" (UXD7)

Avoidance behaviors, such as withdrawing from discussions or meeting only minimum expectations, further exacerbated collaboration challenges. Participants emphasized that fostering psychological safety and trust is essential to addressing these issues.

By creating supportive environments that encourage open communication and mutual respect, teams can mitigate the negative effects of IP and enhance collaborative effectiveness.

4.4 Theme 4: Productivity Impacts of Impostor Phenomenon

Impostor Phenomenon (IP) disrupts productivity by fostering procrastination, overwork, and emotional strain, ultimately hindering collaboration and team efficiency.

A developer shared how self-doubt led them to delay seeking help: "I kind of thought, I shouldn't have to ask for help for this. But instead of asking for help half an hour, and I just left it for like a week." (DEV1)

Similarly, a designer described overworking in an attempt to compensate for self-doubt: "I sometimes stay back, trying to perfect the design even though the changes don't make much difference. I might spend the whole evening doing mental checks." (UXD7)

These behaviors reflect the emotional toll of IP, where fear of failure and perfectionism result in inefficiency, strained resources, and delayed progress. Participants emphasized that fostering open communication and realistic expectations within teams is essential to mitigate these effects. By addressing the root causes of IP, teams can create an environment that supports productivity and collaboration.

4.5 Theme 5: Team Dynamics and Organizational Support for Mitigating Impostor Phenomenon

Strong team dynamics and supportive organisational practices are critical in alleviating the effects of Impostor Phenomenon (IP) on collaboration between UX designers and developers. Participants emphasized the importance of trust, empathetic leadership, and a safe team environment in fostering confidence and improving teamwork.

A developer described how team support helped them navigate IP-related challenges: "But luckily, my team was very understanding as well, and they're looking out for me. And there was like an elite developer who also took some time to make sure that I was okay." (DEV1)

Leadership and management involvement were also seen as vital in addressing IP. Clear communication of expectations and proactive support from managers helped reduce uncertainties and fostered a more collaborative atmosphere. A designer noted: "I don't think this can be solved entirely from the designer's side—it probably requires more support from the product manager or higher up." (UXD7)

Creating an inclusive team culture and implementing supportive management practices, such as regular feedback, clear goal-setting, and open communication channels, were identified as effective ways to mitigate IP. By fostering trust and psychological safety, organisations can enhance team dynamics, enabling designers and developers to contribute more confidently and collaboratively.

5 Discussion

This study reveals how Impostor Phenomenon (IP) affects collaboration between UX designers and developers. The five themes identified highlight key challenges, such as communication barriers, self-doubt, and reduced productivity, while emphasizing the value of cross-disciplinary understanding and supportive team dynamics. These findings offer a basis for designing interventions to address the effects of IP and improve team collaboration.

5.1 The Impact of IP on Collaboration

The findings show that IP exacerbates communication challenges between designers and developers, aligning with prior studies [3, 11, 35], which identified communication as a core challenge in interdisciplinary teams. This study builds on existing research by showing how IP intensifies these challenges through self-doubt and emotional strain. For instance, participants frequently hesitated to share their opinions due to fear of criticism or perceived inadequacy, which supports the findings [46, 47].

Fear of feedback, both in giving and receiving, emerged as another significant issue. Participants often misinterpreted constructive criticism as a judgment on their abilities, leading to decreased trust and collaboration efficiency. This observation aligns with Rudenga's research [47] on feedback aversion linked to IP.

5.2 The Impact of IP on Productivity

The emotional and behavioral consequences of IP negatively affect productivity. Participants reported procrastination as a common response to the insecurity caused by IP, consistent with studies [46, 47, 52]. Overwork and excessive self-review were also common coping mechanisms, with participants spending unnecessary time perfecting tasks to compensate for their perceived inadequacies. This behavior not only reduced personal efficiency but also strained team resources, further illustrating the detrimental impact of IP on workplace performance.

5.3 The Importance of Cross-Disciplinary Understanding

A lack of understanding between designers and developers about each other's roles and processes was identified as a factor intensifying IP. Participants noted that misunderstandings often led to mistrust and collaboration issues. These findings are consistent with studies [17, 39], which highlighted the importance of mutual understanding for effective team dynamics. This study further suggests that initiatives such as role exchanges and cross-disciplinary knowledge sharing could enhance trust, reduce IP's impact, and improve psychological safety in teams.

5.4 The Role of Organizational Support

Supportive team dynamics and organizational culture were seen as critical for mitigating the effects of IP. Participants highlighted the importance of open communication, tolerance for mistakes, and clear role expectations. These observations align with the study [22] on the role of organizational support in reducing IP. Leadership also plays a significant role; participants emphasized the need for active

involvement from managers, including goal-setting and emotional support, to build confidence and foster a collaborative environment.

5.5 Contributions of This Study

This study advances understanding of how Impostor Phenomenon affects collaboration between UX designers and developers, addressing a gap in current literature. By examining the specific effects of IP on communication, productivity, and team dynamics, it provides insights relevant to both academic research and professional practice. Additionally, the study proposes design-driven strategies for mitigating IP's impact, offering organizations practical methods to enhance interdisciplinary collaboration through better design practices.

6 Limitations and Future Works

This study grounded in design thinking, offers insights into how Impostor Phenomenon (IP) affects collaboration between UX designers and developers, but it has limitations.

First, semi-structured interviews, while effective for exploring subjective experiences, are susceptible to self-report bias, with participants potentially tailoring responses to align with perceived researcher expectations.

Second, despite providing a clear definition of IP, participants' varied interpretations may have introduced subjective elements, affecting data consistency.

Lastly, the study focused on the "Empathize" phase of design thinking, emphasizing perceptions rather than directly observing how IP manifests in real-time collaboration, limiting the evidence of IP's impact on team interactions.

Future research will progress to the "Ideate" and "Prototype" phases, using participatory workshops to co-create and test interventions. These workshops will engage designers and developers in collaborative tasks, allowing for real-time observation of IP's effects and the development of practical solutions.

Reflective practices, inspired by Gold et al. [20], will play a key role in fostering self-awareness, trust, and mutual understanding. Observational data from these workshops will complement interview findings, offering a more complete picture of IP's influence on collaboration.

Building on these outcomes, prototypes will be developed and tested in real-world scenarios, following an iterative approach with longitudinal evaluations to assess their effectiveness in mitigating IP and improving team collaboration.

Through participatory workshops and iterative prototyping, this study aims to provide actionable solutions to address IP's challenges, contributing to both academic research and practical strategies for enhancing interdisciplinary collaboration.

7 Conclusion

This study provides important insights into how Impostor Phenomenon (IP) affects collaboration between UX designers and developers. By identifying key challenges such as communication barriers, emotional impacts, and productivity issues, it underscores the need for targeted interventions. The findings form a solid foundation for future design-driven solutions to mitigate IP and enhance team dynamics.

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APPENDICES

A.1 Interview Questions

- What is your background?
- What is your role in your team/organization?
- Could you tell me about how you collaborate with a developer/designer in your work?
- What is the most challenging part of working with a developer/ UX designer?
- If you could improve one thing in your collaboration with others (UX designer or developer), what would it be?
- What work is done as part of a team?
- How do teamwork and communication affect a project in your experience? Could you share a simple example? Why? (Especially for UX design and Development)
- Have you ever heard of impostor syndrome/imposter phenomenon? If so, please tell me about your experiences with it.
- Do you feel that it affects your work and daily life? If so, how? (Impostor Syndrome is a psychological condition where people, despite being successful and competent from an external viewpoint, feel anxious and incapable of owning their success internally.)
- Have you ever discussed IS (Impostor Syndrome) in your work or study?
- How does it affect people in the design/development sector?
- How might it affect collaboration between designers and developers in your opinion?
- Do you have any examples from your experience? What happened there? Why?

A.2 Research Objectives Based on Design Thinking: Own Work

This figure represents the overall research objectives based on design thinking, including future research as a supplementary component.

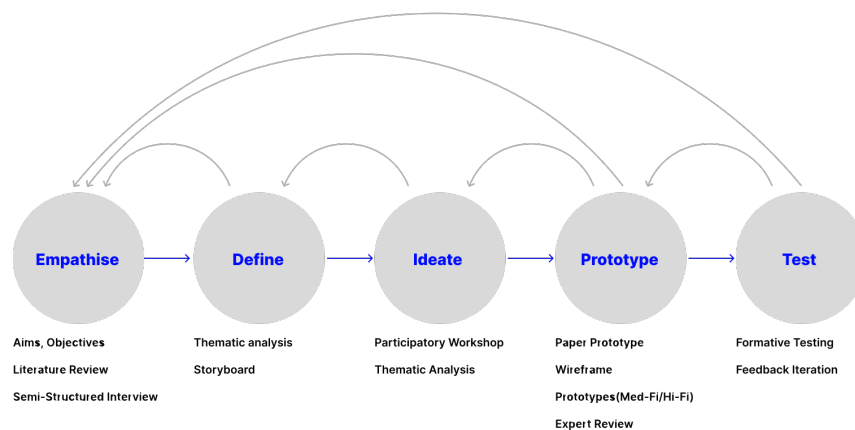


Figure 1: Research Objectives Based on Design Thinking: Own Work